

Richard Xu

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EDUCATION

Western University

London, ON

Bachelor of Engineering Science (BESc), Software Engineering — CGPA: 3.9/4.0

Sept. 2024 – Apr. 2028

Ivey Business School, Honours Business Administration (HBA)

Sept. 2027 – April 2029

Coursework: Data Structures & Algorithms, Software Design, Software Construction, Scripting Language Fundamentals

EXPERIENCE

AI Engineering Intern

May 2026 – Present

Magical

Toronto, ON

- Developed production healthcare automations using **TypeScript, Temporal, LLM agents, Playwright, REST APIs, and Redis/Upstash** across EHR, CRM, Gmail, and Slack systems
- Reduced CSV export latency by **20× (5.5 minutes to 17 seconds)** by replacing nearly 13,000 per-record Redis requests with batched **HMGET** pipelines
- Built Slack-based run observability and defensive API handling, identifying incomplete executions and recovering malformed CRM payloads across 150 production runs
- Eliminated a **6.8% reporting reconciliation gap** across 1,600+ claims through cohort-based deduplication and validation

Software Engineering Intern

Jan. 2026 – April. 2026

Autumn (Startup)

New York, NY

- Built MVP for a full-stack platform to automate pricing and workflow processes for funeral service providers, simplifying complex and manual operations
- Developed backend logic and APIs using **Strapi (Node.js)** and **PostgreSQL** capturing data for over **15,000** funeral houses in the U.S
- Built a responsive frontend using **Next.js** to enable intuitive user interaction and workflow management

AI/ML Research Intern

May 2025 – Aug 2025

Western University

London, ON

- Built an intelligent Wi-Fi optimization platform that dynamically switched access points based on real-time network conditions to improve connectivity stability
- Designed and trained a **PPO reinforcement learning agent** using **Stable-Baselines3** and **Gymnasium**, with a custom environment optimized for latency, jitter, and packet loss
- Engineered a **Flask + React** experimentation platform and benchmarked the agent against static and random strategies across **50+ reproducible episodes per run**

PROJECTS

Slack Digest AI | *Javascript, Node.js, Express, Slack API, REST APIs*

April 2026

- Built a fullstack automation tool that aggregates and summarizes Slack messages into structured daily digests
- Integrated the **Slack API** to fetch channel data and process conversations in real-time
- Developed backend endpoints to handle message processing and digest generation
- Designed a simple frontend interface to display and interact with generated summaries

eBay Deal Finder | *Python, Flask, React, eBay API, PostgreSQL, Render*

July 2025 – Aug 2025

- Built a full-stack web app that lets users search for products and find real-time deals using the **eBay API**
- Integrated eBay production and sandbox APIs, handling **authentication, queries, and dynamic responses**
- Implemented product sorting by price/rating and stored search history

TECHNICAL SKILLS

Languages: Python, Java, Javascript, HTML/CSS, C#, Typescript

AI / ML: LLMs (API-based), Prompt Engineering, RAG, Agentic Systems, Reinforcement Learning

Frameworks: React, Next.js, Flask, Django, FastAPI, Strapi (Node.js), .NET, Unity, Playwright

Libraries/Databases: pandas, NumPy, Matplotlib, PostgreSQL, Redis/Upstash